Reading dossier: *Diagnose or Retrain?*

Date: 6 Oct 2026 · **Status:** Direction adopted 6 Oct; design not frozen; kill-test Stage 0 running on 6 Oct (no result yet).

1. What we are doing

The question. When benign fine-tuning makes an aligned model less safe, which data fix restores safety without hurting the task or causing over-refusal? How much diagnosis is worth paying for before we choose?

- **Fixes:** keep the data, remove a source, add safety data, rewrite or self-distill.
- **Diagnostics:** classifier scores, representation similarity, training-data attribution, short trial runs.
- **Setting:** Qwen2.5-1.5B-Instruct with LoRA. Main: benign erosion (Dolly). Anchor: overt injected harm, rebuilt on the same model.

Why it matters. Fine-tuning services can block an unsafe fine-tune and tell the customer to fix the data and resubmit (one documented case; grant dossier §1a). Fixes have side effects, and we found no tested, costed rule for the choice.

What is new: only the combination. Each part exists alone. Closest: Xiao and Aranguri (no trial-run rival, no add fix, no seeds); SafeTune (no data fixes, one seed); DataRx (no removal, no costs).

Evidence so far (development only).

- **v2** (base model, injected unsafe answers, 2 seeds): a classifier-targeted rewrite cut sampled unsafe answers from 80.3% to 11.4%. But XSTest over-refusal rose from 1% to 44%.
- **Round 1** (2 seeds): a fixed-size top-K removal rule (1,015 rows; 43 harmful-answer rows in the editable pool) raised unsafe answers from 12.5 to 24.0 of 72.

Main measures. (1) Decision value per diagnostic against trial runs at matched compute, as a curve over the cost ratio (diagnostic cost ÷ one full run). (2) Side-effect prediction: does a diagnostic flag the fixes that break the over-refusal or task-skill limit better than trial runs of equal cost?

2. Reading list (22 core papers)

★ **= read first** (the first six starred papers, in the list's own order). **Full** = read in full through a fetch tool that summarizes pages; numbers are "reported" until checked in the PDF. **Summary** = abstract, metadata or earlier notes only; UNVERIFIED.

Paper	arXiv	Why we read it	Status
A. Closest work			
★ Xiao and Aranguri	2602.11079	Prices several diagnoses; applies three data fixes	Summary (abstract checked)
★ SafeTune (Seth et al.)	2609.22153	Likeliest "this was done" citation; model-level repairs only	Full
★ DataShield (Zhang et al.)	2606.00160	Benign-erosion filter; prices scoring, never against runs	Full
★ DataRx (Zhang et al.)	2608.04322	Which safety rows to add; measures the over-refusal cost	Full
★ GradShield (Hu et al.)	2605.14194	Threshold set by safety and utility tolerances: nearest precedent for our limits	Full
Luo et al.	2609.02771	Rewriting beats deletion; the rewrite fix is not ours to claim	Full (HTML; 5 Oct note)
Pan et al.	2502.11411	Attribution beats classifiers on injected rows, from one run; we replicate it with seeds	Summary
B. Safety erosion after benign fine-tuning			
★ Qi et al. 2023	2310.03693	The founding result (Alpaca, Dolly); defines the main setting	Summary (tool summary; read in full before citing)
Kim et al.	2508.12531	Better hyperparameters alone cut unsafe answers from 16% to about 5%	Summary
Lyu et al. (PTST)	2402.18540	Prompt templates decide much of the erosion; PTST is a near-free non-data repair	Summary
C. Finding the cause: classifiers, representations, attribution			
He, Xia and Henderson	2404.01099	Selected benign rows break safety; never build data with the same signal	Summary (tool summary)
DATE-LM (Jiao et al.)	2507.09424	With look-alike data, attribution detection drops while classifiers hold	Summary (one passage checked)
Persona vectors (Chen et al.)	2507.21509	Flags data that will shift a trait: nearest precedent for side-effect prediction	Summary
D. Fixes: add, rewrite, self-distill			
Bianchi et al.	2309.07875	About 3% safety data helps; too much causes over-refusal	Summary
Eiras et al.	2406.10288	Safety data in the task's format: the strongest add arm	Summary
SDFT (Yang et al.)	2402.13669	The baseline self-distill fix	Summary
López-Ávila et al.	2609.04482	Refusal data raised XSTest over-refusal from 2% to 74%	Summary
Dubiński et al.	2604.25891	Added data can hide misalignment that returns on training-like prompts	Summary

E. Decisions under a compute budget			
DataDecide (Magnusson et al.)	2504.11393	Small runs scored by decision accuracy against compute	Summary
Ivison et al.	2503.01807	Charges selection compute against more training data	Summary
F. Evaluation and judges			
Harmful fine-tuning survey (Huang et al.)	2409.18169	Field practice: no over-refusal, no seeds, cost only as future work	Full (HTML read as text)
Vishnubhotla et al.	2606.03648	Judges misread incoherent outputs; compare fixes at fixed task skill	Full

Full list: 134 entries, with URLs in reading order, in `research/review/2026-10-06-reading-list-v2.md`. Cite the two "DataShield" papers by ID (2606.00160 and 2607.15081).

3. Five stages to a complete study

E = rise in the unsafe rate over the untuned model, in percentage points (pp).

Stage	Goal	Main work	Gate or stop rule	Cost (estimate)	Weeks and dates	Output
1. Kill test	Show that erosion exists, is localized, and that attribution finds it	Stage 0 now; Dolly and Alpaca × 3 seeds; random-removal null; one removal run per source, then repeats; early side-effect check (3% safety mix)	Futility: E < 3 pp after one seed. Gate A: E ≥ 10 pp (15 to go on). Gate B: diffuse → kill source-level diagnosis. Gate C: attribution must beat representation similarity beyond seed noise, else shrink	$127–145 (first pod $5.22)	Weeks 1–2 (6–19 Oct)	Go, shrink or kill
2. Build, calibrate, design freeze	Build the study; fix its design	Sources as units; add and self-distill fixes; task-skill and coherence checks; representation similarity; costs in full-run units; successive halving; anchor rebuilt on the aligned model; calibration runs	Old recipes stay byte-identical. Design freeze hashed before development outcomes	$100–300	Weeks 2–4 (13 Oct–2 Nov)	Engine; frozen design
3. Development, numerical freeze	Measure every arm; fix every number	2 settings × ≥ 5 seeds; default add arm (diagnosis must beat it); Pan replication with seeds	Pre-set margins: each gain is negative, positive or inconclusive (→ new data or stop). Numerical freeze before hidden access	$1,500–3,000	Weeks 5–9 (3 Nov–7 Dec)	Development tables; frozen numbers
4. Held-out confirmation	Test the frozen claims on unseen data	Held-out menus with sealed contamination; fresh seeds	No retuning; nulls and near-ties are reported	$1,000–2,000	Weeks 10–11 (8–21 Dec)	Confirmatory results
5. Analysis, paper, release	Report and release	Analysis; writing; code, protocols, cost tables	ICML needs about $3–5k by early November; else NeurIPS 2027 (about mid-May; UNVERIFIED)	No GPU line	Weeks 12–15 (22 Dec–18 Jan); ICML deadline about 22 Jan 2027 (UNVERIFIED)	Paper, nulls included; open code

Dates come from the BlueDot draft §5. Its single week-9 freeze is now the numerical freeze; the design freeze ends stage 2 (the user's two-freeze decision).

- **Money.** The user chose a $6,000 BlueDot ask on 6 Oct; it is not yet submitted. It covers all five stages (about $4,700 of it is GPU work, with contingency). The RunPod balance covers Stage 0 only, and every later spend needs a new approval.
- **What kills or narrows the study.** No erosion after the fallback ladder kills this setting at this scale. Diffuse erosion narrows the study to adding data. If attribution is no better than representation similarity, we drop or demote it. If no diagnostic beats random removal, we report "retrain wins". If the default add arm restores ≥ 80% of the erosion within +5 pp over-refusal, the default fix is enough here. We report every null.
- **Open decisions with the user.** FOCUS revision draft §15, items 2–9 (how decision value and side-effect prediction are defined and read, "remove a subset", the cost unit, non-data repairs) and §16 item 10 (how to test "beat the default mix"); kill-test spend after the first pod, Alpaca and the row-level removal arms; the BlueDot fields only the user can fill.

4. Claim rules

> - Never claim that practitioners ask whether attribution is worth its compute. We found no public evidence that they do.
> - Describe round 1 only as a finding about a fixed-size top-K removal rule, never as "classifiers fail".
> - Novelty is the combination. Do not claim rewriting, attribution-guided selection or "attribution beats classifiers" as new.
> - Every result so far is development evidence. Nothing is confirmatory.

Sources: research/FOCUS.md; research/review/2026-10-06-reframing-proposal.md, -reading-list-v2.md, -focus-revision-DRAFT.md, -prior-work-search.md, -positioning-review.md; research/experiments/2026-10-06-kill-test-design-DRAFT.md, 2026-10-06-headroom-v2-results.md, 2026-10-05-headroom-round-1-results.md; .ai/briefs/kill-test-stage0-engine-20261006.md; research/grant/2026-10-06-bluedot-application-DRAFT.md.